

OSBAMS Rev.2 controller PCB — assembly notes

Purpose. OSBAMS Rev.2 Controller PCB is a battery measurement and safety-control board. The PCB does not carry the main battery discharge current. High-current battery connections are external.

STATUS: NOT RELEASED. Do not start assembly until the HOLD items below are cleared in writing by the design owner.

Important assembly rules

- **Do not substitute the STM32** (NUCLEO-L476RG / STM32L476RG) without approval — it is a plug-in module, not assembled by the fab.
- **Do not substitute the INA228** (Adafruit 5832 breakout) without approval — plug-in module, not assembled by the fab.
- **Maintain the measurement components:** U1 TC74A5-3.3VAT (Microchip) must not be replaced by another sensor or address variant.
- **Safety parts — no substitution without approval:** Q1 IRLZ44NPBF, R2 gate pull-down, D1/D2/D3 (once polarity is resolved).
- **Verify connector orientation:** J1–J3 (terminal blocks, wire entry toward the board edge as designed), J4/J5/J6 pin 1 at the square pad. J5 is not physically keyed.
- **Verify the programming header:** there is NO programming header on this PCB; programming is through the ST-LINK on the Nucleo module. Do not fit anything in J5/J6 other than the intended modules.
- **Verify polarity-sensitive parts:** C1 (+ on pad 1), D1/D2/D3 (cathode band; **HOLD — schematic and footprint disagree**), Q1 (G-D-S), U1 (pin 1 NC, square pad).
- **Do not fit** the battery fuse, XT60 power connector, disconnect, relay contacts, shunt wiring or the 6060B connection — external power hardware.
- No battery or high-energy source is to be connected during factory work.

Part-by-part notes

Ref	Part	Mount	Orientation / note
C1	10 uF, 25 V / ECA-1EM100I	THT	THT. Polarised: + lead = pad 1 = +12V. MPN comes from the schematic value. 25 V rating vs SMBJ15A clamp (~24 V peak) leaves little margin - review item.
C2, C3	0.1 uF (100 nF) / NOT SPECIFIED	THT	THT. Voltage rating and dielectric not specified (needs >= 16 V, X7R/Z5U class) - customer must pick an MPN. Not polarised.
D1	SMBJ15A / SMBJ15A	SMT	ONLY SMT PART ON THE BOARD. Polarity: schematic puts cathode on +12V; KiCad D_SMB footprint pad 1 (left, cathode) is on GND - POLARITY MISMATCH, resolve before ordering (see review). Manufacturer not named in design (Littelfuse/Vishay/Bourns all make SMBJ15A) - needs approval.
D2	1N5408 / 1N5408	THT	THT. Cathode band = pad 1 (square pad, silk 'K'). Schematic wants cathode on COIL_V (+12V side) but pad 1 is on COIL_SW - POLARITY MISMATCH. Manufacturer not named (e.g. Vishay 1N5408-E3/54 would be a candidate - needs approval).
D3	SB560-E3/73 / SB560-E3/73	THT	THT. MPN from the schematic value. Cathode band = pad 1, which the PCB connects to 12V_IN; schematic intends cathode on +12V (load side) - POLARITY MISMATCH, resolve before ordering.
Q1	IRLZ44NPBF / IRLZ44NPBF	THT	THT. Pinout G-D-S = pads 1-2-3, matches the footprint. Gate driven from a 3.3 V GPIO via R1; do not substitute (safety part). No heat sink fitted/specified.
R1	220 ohm / NOT SPECIFIED	THT	THT. Tolerance/power/series not specified (assumed 1%, 0.25 W) - customer must pick an MPN.
R2	10 kohm / NOT SPECIFIED	THT	THT. Holds the gate low if the MCU pin floats (safety-relevant). Tolerance/power/series not specified - customer must pick an MPN.
U1	TC74A5-3.3VAT / TC74A5-3.3VAT	THT	THT. Measurement part - do not substitute. NOTE the older repo BOM lists TC74A0-5.0VAT (5 V, addr 0x48); this PCB/schematic use the 3.3 V A5 variant - confirm. Pin 1 is NC. Pads have 0.087 mm annular ring (< 0.15 mm).
J1, J2, J3	MX126-5.0-02P (2-pos, 5.0 mm) / NOT SPECIFIED	THT	THT. Footprint name suggests 'MaiXu MX126-5.0-02P' but no MPN/manufacture is in the schematic. These carry the 12 V coil current only (NOT the battery discharge current). No +/- legend on silkscreen.
J4	INA228_MODULE (1x04, 2.54 mm) / NOT SPECIFIED	THT	THT. The INA228 itself is NOT on this PCB (Adafruit 5832 breakout plugs in; not assembled by the fab). Header MPN not specified.
J5	NUCLEO_INTERFACE (2x04, 2.54 mm) / NOT SPECIFIED	THT	THT. The STM32L476RG is NOT on this PCB (it is on the Nucleo module). Schematic note says 'keyed' but a plain header footprint is not keyed - orientation risk.
J6	UART_HDR (1x04, 2.54 mm) / NOT SPECIFIED	THT	THT. Footprint is a fan-style header with locating pegs (two 1.1 mm NPTH); no MPN or mating connector specified - confirm part vs footprint.

Process notes

- Order mix: 1 SMT part (D1) on the top side; all other parts through-hole, top side only. No bottom-side components.
- TO-220 parts (Q1, U1) mount vertically with the leads through the board; no heat sink or insulation hardware is specified.
- Terminal blocks carry the 12 V coil current only; wire gauge by the user.
- Cleaning: no-clean flux acceptable; avoid flux residue on headers J4/J5/J6.
- Mounting holes MH1–MH4 are 3.2 mm non-plated; keep mask/paste out of them.
- Silkscreen has designators only — no polarity legend on J1–J3; refer to the Assembly Drawing pin/polarity map.